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Paleomagnetic and rock magnetic study of the Vredefort impact structure and the Johannesburg Dome, Kaapvaal Craton, South Africa—Implications for the apparent polar wander path of the Kaapvaal Craton during the Mesoproterozoic

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ABSTRACT

New paleomagnetic and rock magnetic results are presented for various pre-, syn- and post-impact lithologies from the region of the 2023 ± 4 Ma, originally ca. 250 km diameter, Vredefort impact structure, on the Kaapvaal Craton of South Africa. After removal of a viscous remanent magnetization (VRM) component, the characteristic remanent magnetization (ChRM) component from 16 Vredefort Granophyre and pseudotachylitic (PT) breccia samples ($D = 18.3°$, $I = 54.8°$, $\alpha_{95} = 8.1°$) was isolated and a paleomagnetic pole at 25.1°N and 43.5°E ($A_{95} = 10.6°$) obtained. Since, such a ChRM component was isolated also from the Archean Basement rocks of the Vredefort Dome, we consider its nature as a primary component of the impactites proven. The pole falls onto the Paleoproterozoic part of the apparent polar wander path of the Kaapvaal Craton.

Rock magnetic analysis and scanning electron microscopy (SEM) revealed that the pseudotachylitic breccia and Granophyre samples contain two distinct magnetite phases: one ultra-small (micrometer-size), and one altered and larger (>50 μm) phase. The ultra-small magnetite is interpreted as the carrier of the ChRM and the larger grains as the carriers of the VRM component.

Additionally, rock magnetic and petrophysical data reveal unusually high Koenigsberger ratios (*Q* values) in all pre-impact lithologies, in some Vredefort impactite samples, and in the much younger (1.1 Ga) Anna's Rust Sheet (ARS) gabbro samples. As the high *Q* values, which had also been reported by previous studies of Vredefort lithologies, are now also seen in samples from the Johannesburg Dome, a direct link to the Vredefort impact can be ruled out. This is also supported by the hysteresis data of this study. As the observed magnetization is rather hard and shows multiple components of remanent magnetization, we exclude lightning as a cause for all observed high *Q* values (except in case of ARS gabbros). It is instead suggested that the cause of the high *Q* values could be related to the high temperatures of the rocks that were uplifted by the impact event from a mid-crustal original setting, and to fluid circulation within the two domes that made the rocks vulnerable to acquire high thermochemical remanence.

Moreover, paleomagnetic analysis of the rocks around the ARS gabbro intrusion in the northern part of the Vredefort structure revealed the presence of either a shallow north or a shallow south direction, which is tentatively related to emplacement of the Umkondo large igneous province. Analysis of all rocks, including the Vredefort impactites, yields occasionally distinct great circle paths towards these shallow directions. A likely explanation for this overprint direction is the heating caused by now eroded ARS-type gabbro in the area, or by regional, ca. 1.1–1.0 Ga orogenic effects. No evidence of Karoo-type (0.18 Ga) overprint is seen.

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1. Introduction

Paleomagnetic studies of reliably dated rocks from stable cratonic regions are fundamental for understanding ancient plate motions. The Kaapvaal (South Africa) and Pilbara (Australia) cratons are the only two extensive mid-Archean crustal fragments on

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Earth. The Kaapvaal Craton represents the Archean basement to the north-central part of the southern African subcontinent. The proto-Kaapvaal Craton formed over a period of about 500 million years, between 3.6 and 3.1 Ga, through amalgamation of small tectonic blocks (e.g., Eglington and Armstrong, 2004; Schmitz et al., 2004). Stabilization of the entire craton probably occurred around 2.9 Ga (e.g., de Wit et al., 1992). More extensive cratonic entities that became stabilized following Mesoproterozoic orogens in Africa are the Congo and Kalahari Cratons. In southern Africa, these cratons are separated by a transcontinental Pan-African orogen comprising the Damara Belt, the Lufilian Arc, and the Zambezi Belt (Hanson, 2003).

The stable Kaapvaal Craton experienced a massive meteorite impact event at 2023 ± 4 Ma (2σ) ago (Kamo et al., 1996), which resulted in the Vredefort impact structure. This structure is the oldest and, with an estimated original diameter of 250–300 km (Henkel and Reimold, 1998), the largest impact structure identified on Earth (for a review of the Vredefort impact structure see Gibson and Reimold, 2001, 2008).

Rocks from the Kaapvaal Craton are good candidates for providing paleomagnetic data for the study of the continental drift of the cratons in the period between the time interval for the Kenorland supercontinent (2.45–2.1 Ga) and formation of the Early Proterozoic Hudsonland supercontinent (1.83 to 1.50–1.25 Ga; also known as Columbia or Nuna, e.g., Condie, 2004; Karlström et al., 2001; Meert, 2002; Pesonen et al., 2003; Rogers and Santosh, 2002; Zhao et al., 2003, 2004). The period 2.4–2.1 Ga documents important geological and biochemical changes on Earth, such as δ^{13}C-excursions, onset of black shale and phosphate deposition, and worldwide glaciations (e.g., Melezhik et al., 2005, and references therein). Moreover, there was a radical change in biogeochemical recycling of organic matter at ca. 2060 Ma (Melezhik et al., 2005). Paleomagnetic data from the impact-generated rocks of the Vredefort impact structure may provide a tool to study the interval between these two supercontinents. However, earlier paleomagnetic studies of Vredefort lithologies by Hart et al. (1995, 2000) and Carporzen et al. (2005) (see also petrophysical results by Henkel and Reimold (1996, 1998, 2002)) have revealed the problematic nature of these Archean rocks, namely the extraordinarily high values of natural remanent magnetization (NRM). We have undertaken a detailed paleomagnetic and rock magnetic study of geologically and chronologically well constrained samples from the Vredefort impact structure and extending the sampling region to the Johannesburg Dome, in order to further elucidate these problematic issues and to attempt to evaluate the paleomagnetic potential of this region for paleoreconstructions.

This study also contributes to the controversial discussion about suggested temperatures in the core of the Vredefort structure before and after the impact event (Carporzen et al., 2006; Gibson and Reimold, 2005; Gibson et al., 1997; Henkel and Reimold, 2002; Turtle et al., 2003; Muundjua et al., 2007). Textural evidence indicates that immediately after the passage of the impact-generated shock wave the maximum temperatures experienced by rocks varied along the radius of the Vredefort Dome (Gibson and Reimold, 2005; Gibson et al., 1997). There is mineralogical evidence that the temperatures of the rocks currently exposed in the central core (radius $\leq$ 8 km) of the Vredefort Dome exceeded the minimum granite solidus (~650–700 °C; Gibson, 2002; Gibson et al., 1997). Turtle et al. (2003) and Ivanov (2005) modeled similar temperatures in the core region, with maximum temperatures in the center of the dome exceeding 900–1000 °C. Moreover, according to Ivanov (2005) the wider central part of the core (radius $\leq$ 15 km) was heated above 700 °C. In contrast, Carporzen et al. (2006) suggested that temperatures of the core rocks never rose above 550–600 °C for 3 min, or above 500 °C for 1 h. However, this is unlikely given the nature of metamorphic processes after the very large Vredefort impact event: post-impact heat took several hundreds of thousands of years to dissipate, as modeled by Ivanov (2005) and Ivanov and Deutsch (1999).

A series of pre-impact alkaline and mafic intrusive complexes occur in and around the outer reaches of the Vredefort Dome. With this paleomagnetic study we also contribute to the discussion raised by Hargraves (1970, 1987) concerning the timing of emplacement of the Rietfontein, Roodekraal, Schurwedraai and Lindeques Drift intrusions relative to the overturning of the supracrustal strata occurring in the outer parts of the Vredefort Dome.

Finally, we have investigated whether the samples from various lithologies show any overprinting from younger geological events, such as the emplacement of the Anna's Rust Sheet (ARS) gabbro (ca.1.1 Ga), events related to the emplacement of the Umkondo large igneous province at ca. 1.1 Ga or to the accretion of the Namaqua-Natal Belt (ca. 1.1 Ga) onto the Kaapvaal Craton, or to the emplacement of Karoo basalts (0.185 Ga).

2. The regional setting of the Vredefort Dome and sampling sites

Reimold and Gibson (1996) and Gibson and Reimold (2001, 2008) have reviewed the geology of the Vredefort impact structure. The extent of the eroded remnant of this impact structure encompasses all of the Witwatersrand Basin, stretching for 300 km between Johannesburg in the northeast to the Welkom goldfields in the southwest. Located roughly in the central part of the basin, the Vredefort Dome represents the ca. 90 km wide, exhumed root of the central uplift of the Vredefort impact structure. Fig. 1 is a schematic geological map showing the simplified geology of the Vredefort Dome, and also our sampling sites. The southern and eastern parts of the Dome are generally overlain by sedimentary strata and dolerites belonging to the Karoo Supergroup (185 Ma).

At the northeastern edge of the Witwatersrand basin lies the Johannesburg Dome, an Archean terrain composed of granitoids and greenstone remnants with several inliers of Witwatersrand Supergroup (WSG) strata (Fig. 2, after C.R. Anhaeusser, pers. comm.). Immediately north of the Johannesburg Dome, the rocks of the 2.6–2.15 Ga Transvaal Supergroup (TSG) dip shallowly to the north beneath the 2.05–2.06 Ga old Bushveld Complex (Cawthorn et al., 2006; Eriksson et al., 2006; Reimold and Gibson, 1996).

Our main sampling was done in the exhumed mid to upper crustal lithologies of the Vredefort Dome. An additional 12 samples of Archean basement complex (ABC) as well as 4 WSG shale samples from the Johannesburg Dome were taken for comparison, as this terrain was not as strongly thermally overprinted by the impact event as the rocks of the Vredefort central uplift.

2.1. Target rocks from the Vredefort Dome

The Vredefort Dome, centered at 27°S and 27°30′E, is located about 120 km southwest of Johannesburg. It comprises a ca. 40 km wide core of Paleo- to Meso-Archean granitoids (granites, granitic gneisses and migmatites) that is surrounded in the west, north and east sectors by a 15–20 km wide collar of up- and over-turned Meso-Archean to Paleoproterozoic supracrustal strata (Reimold and Gibson, 1996). Thirty-six samples of the ABC from the central part of the structure – the so-called core – were collected.

The WSG was formed between ca. 2.97 and ca. 2.71 Ga ago, when an up to 7-km thick sequence of clastic sediments was deposited (Robb et al., 1997). Nine oriented WSG samples (quartzites, metapelites and metashales) were taken for this study. The eruption of up to 3 km of tholeiitic flood basalts of the Ventersdorp Super-

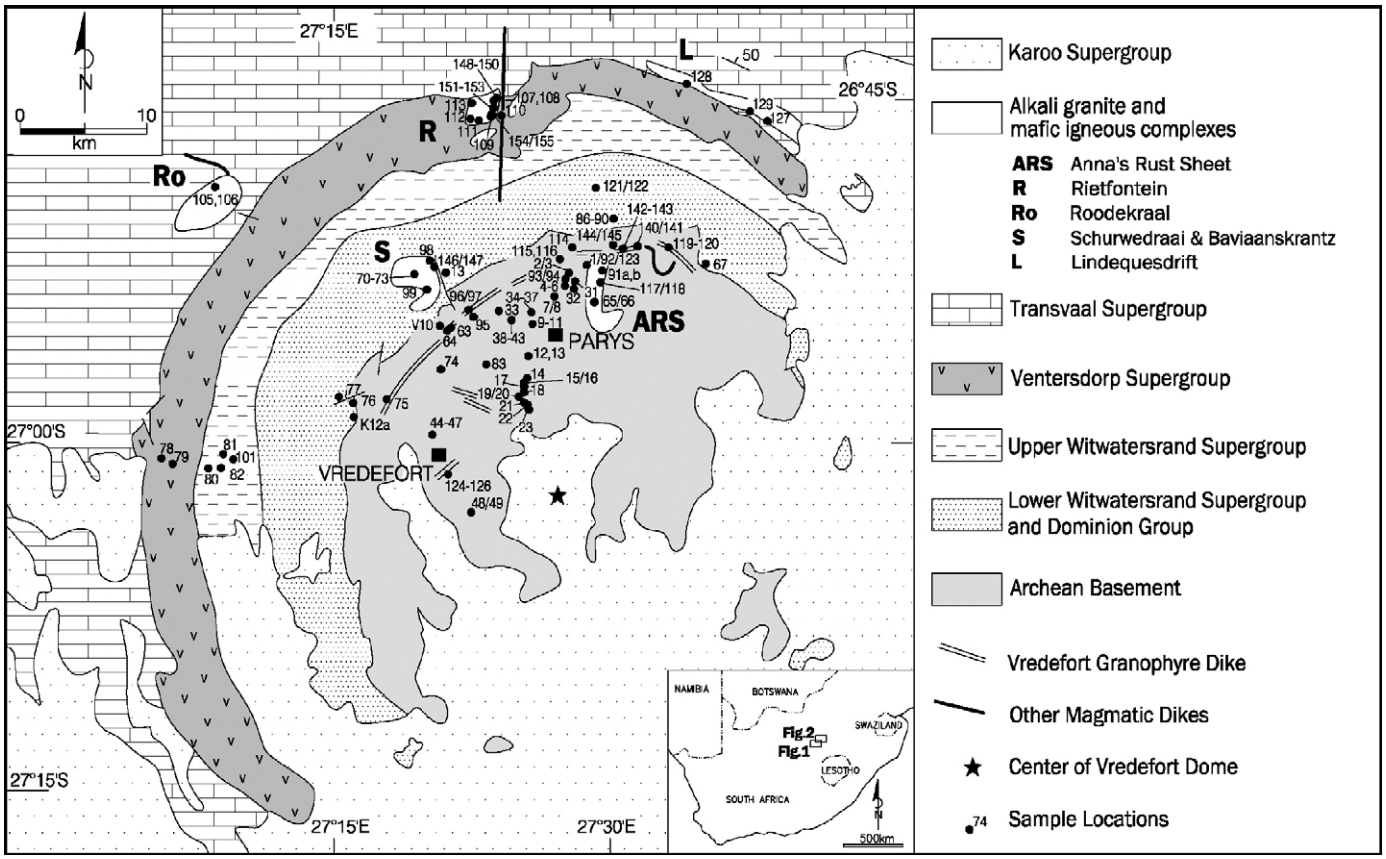


Fig. 1. Simplified geology of the Vredefort Dome.

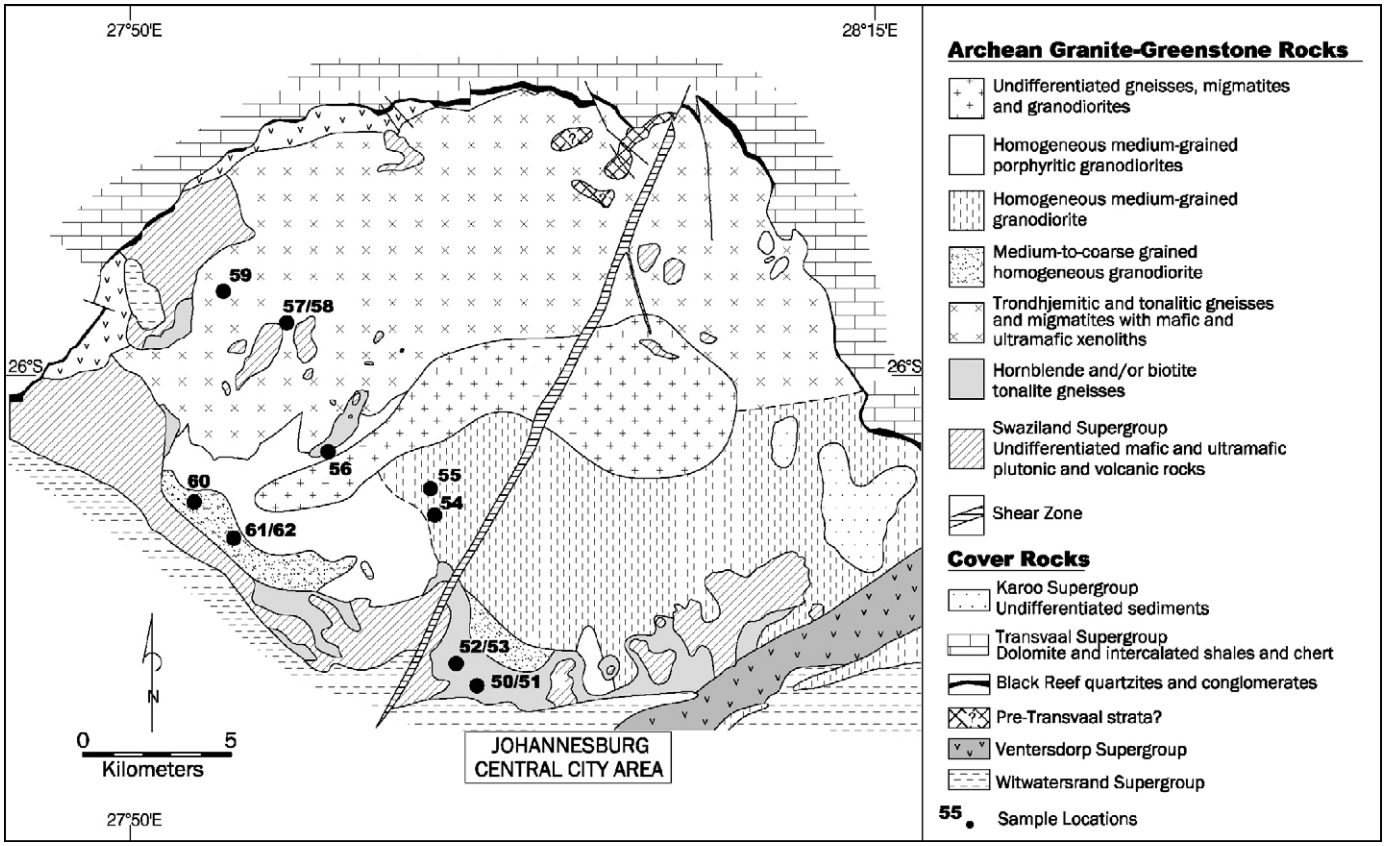


Fig. 2. Simplified geology of the Johannesburg Dome. Sampling sites are denoted by dots.

group (VSG) terminated Witwatersrand sedimentation at 2.714 Ga (Armstrong et al., 1991). Four VSG lava samples were analysed. Formation of the VSG was followed by TSG sedimentation, which commenced at ca. 2.6 Ga ago and likely lasted until about 2.15 Ga ago (Frimmel et al., 2005; Eriksson et al., 2006). At that time much of the Kaapvaal Craton was covered by a shallow sea and up to 2 km of dolomite and 3–4 km of clastic sediments were deposited (Walraven and Martini, 1995). Four TSG samples were collected for this study.

In the inner collar of the Vredefort Dome, the stratigraphy comprises up to 30% mafic intrusions (Reimold and Gibson, 1996). Owing to their metamorphosed and altered state, these dioritic–gabbroic intrusions have been named epidiorites (Bisschoff, 1988; Reimold and Gibson, 1996). The emplacement age of these epidiorites is not known, however, they have in the past been considered as sills related to the possible feeder system for the Ventersdorp Supergroup eruptions (Pybus, 1995; Reimold et al., 2000). Fourteen epidiorite samples were analysed.

2.2. Pre-impact intrusive bodies

Limited sampling was also carried out in the four alkali-granite and mafic intrusive bodies of the Vredefort Dome region (white in Fig. 1) to investigate whether they were intruded before or after the overturning of the Vredefort structure due to the impact event (see Hargraves, 1970, 1987; see also Verwoerd, 2006). Recent dating (de Waal et al., 2006; Graham et al., 2005) of some of these intrusions has yielded pre-impact ages 2053–2055 Ga.

Eleven samples from the Rietfontein Complex were taken. This complex, which comprises a olivine gabbro (samples LV109 and LV154-155) and of alkali granite (samples LV151-153; see also Hargraves, 1970), is intruded into the TSG. This complex is most probably related petrogenetically to the other small, alkaline intrusions within the collar, some of which have yielded zircon ages of ca. 2055 Ga (Graham et al., 2005). The complex is transected by a large nepheline syenite dike (samples LV107-108 and LV148-150) of indeterminate, but clearly post-impact origin.

The Roodekraal Complex consists of various diorites and andesite intruded into the TSG. It is located to the northwest of the Vredefort Dome. These lithologies are cut by a syenite dike that, according to Gadise (1996), is cut by pseudotachylitic (PT) breccia related to the Vredefort impact event, which would thus indicate a pre-Vredefort age for the mafic complex. Moreover, de Waal et al. (2006) reported a SHRIMP zircon age of 2053 ± 9.2 Ma for these diorites. Two samples (samples LV105 and LV106) were taken from this intrusion.

The Lindeques Drift intrusion, which is considered to be related to the same petrologic suite as Roodekraal, was also sampled (samples LV128 and LV129). This body of elongate geometry is composed of diorite. It occurs in the north-northeastern sector of the outer collar of the Vredefort Dome (Fig. 1). The intrusion is located at about the same stratigraphic position as the Rietfontein Complex, within lower TSG metasediments (Hargraves, 1987). de Waal et al. (2006) published a zircon SHRIMP age of 2054.8 ± 5.7 Ma for this intrusion.

The Schurwedraai alkali granite body is intruded into the Lower WSG in the inner north-northwest sector of the collar of the Vredefort Dome. It comprises both alkali granite and nepheline syenite (Walraven and Elsenbroek, 1991). Both lithologies are extensively transected by pseudotachylitic breccia veining, and they also display shatter cones: clear evidence for a pre-impact emplacement of this body. Graham et al. (2005) published a zircon age of ca. 2052 Ma for the Schurwedraai alkali granite. Five samples of this body were analysed (LV70, LV72-73, LV99, LV146).

2.3. Impact-related lithologies

Occurrences of two distinctive melt breccia lithologies are widespread in the Vredefort Dome. Veins and dikes of less than a centimeter to >10 m thickness and up to kilometers in length of pseudotachylitic breccia occur throughout the Archean Basement Complex of the Vredefort Dome and, to a lesser degree, in the other pre-impact rocks as well (Reimold and Gibson, 2005). The processes at generation of these melt breccias have been subject of considerable debate, but the existence of some veinlets with coesite and stishovite (Martini, 1978, 1991; White, 1993) shows that there is at least a component of shock-melted material amongst them. Evidence for a friction-melting component is rare (Dressler and Reimold, 2001), and how the massive volumes of such melt breccia have been emplaced is still subject to multiple research projects.

A series of large dikes of so-called Vredefort Granophyre occur within the ABC and along the core–collar boundary of the Vredefort Dome. Since verification of the presence of a geochemical meteoritic component in the Granophyre (Koeberl et al., 1996) and, finally, of shock metamorphosed minerals and rock clasts (Buchanan and Reimold, 2002), this lithology has been confirmed as the last remnant of the Vredefort impact melt (Reimold and Gibson, 2005, and references therein). Several zircon chronological investigations using TIMS and SIMS (Kamo et al., 1996; Gibson et al., 1997) and argon chronology (Spray et al., 1995) have established a 2020 ± 5 Ma average age for the Vredefort impact event; however, generally the 2023 ± 4 Ma result by Kamo et al. (1996) has been quoted as the impact age. Fifteen PT breccia samples and fourteen Granophyre samples were taken for the determination of the impact-related remanent magnetization direction.

2.4. Post-impact intrusive body

The core of the Vredefort Dome exhibits large occurrences of about 100 m thick, subhorizontal gabbro body known as the ARS, which has been dated at 1054 ± 13 Ma (Rb–Sr biotite and plagioclase age; Reimold et al., 2000; see also Coetzee et al., 2006) and more recently at 1108.6 ± 1.2 Ma (U–Pb baddeleyite age; Hanson et al., 2004a,b). This post-Vredefort impact intrusion age is consistent with the complete absence of evidence of Vredefort-related deformation. Six ARS gabbro samples were collected (LV65-66, LV91a, LV117-118, LV123). Additional samples of ABC lithologies (LV92-94) were collected close to the ARS in order to study the possible baking affects of this intrusion.

3. Methods

Hand samples were collected and oriented in the field using a magnetic compass. As in the laboratory several cores were drilled out of each hand specimen, such samples correspond to what would be called a separate site in other studies. All physical measurements were carried out at the Solid Earth Geophysics Laboratory of the University of Helsinki and at the paleomagnetism laboratory of the Geological Survey of Finland. Petrophysical properties (susceptibility, intensity of NRM, and density) were measured on each specimen before paleomagnetic measurements. Alternating field (AF) demagnetizations were done using the 2G-DC SQUID magnetometer (maximum field up to 160 mT) and AGICO-LDA-3 AF demagnetizer (maximum field up to 100 mT). Stepwise thermal demagnetizations (temperatures up to 680 °C) were carried out using a Schonstedt TSD-1 furnace. During thermal cleaning the susceptibility was monitored after each step in order to detect mineralogical changes due to heating.

The nature of the magnetic carriers was studied by thermomagnetic analysis of selected specimens using the Agico's CS3-KLY-3S Kappabridge system, which measures the bulk susceptibility (k) of the samples during heating up to 700 °C (in argon gas) and cooling back to room temperature. Some of these specimens were also cooled from room temperature down to 80 K and then heated back up to room temperature in order to investigate the possibility of low-temperature transitions of magnetic minerals (e.g., Verwey transitions for magnetite). For determination of domain states of the magnetic carriers, hysteresis properties were measured using a Princeton Measurement Corporation's MicroMag^{TM}3900 model Vibrating Sample Magnetometer (VSM). The origin of the magnetization and grain size distributions of selected specimens were studied further using the Lowrie–Fuller (Lowrie and Fuller, 1971) test. In the case of only one population of magnetic carriers this test may also be used to provide evidence for the origin of the magnetizing field. If the magnetization on the rocks is dominated by thermoremanent magnetization (TRM), the NRM decay should be similar to that of ARM (e.g., Lowrie and Fuller, 1971; O'Reilly, 1984). However, if the magnetization is caused by lightning or other high fields (e.g., a plasma field as proposed by Carporzen et al. (2005)) then the NRM decay should be similar to the decay of IRM (O'Reilly, 1984). Especially lightning is known to produce strong but very soft (IRM) remanences with high Q values (e.g., Irving, 1964).

To discriminate between single domain (SD), multi-domain (MD), bimodal (BM) and mixed behaviors, the median destructive field (MDF) was calculated as the demagnetizing field required to halve the magnetization. Dunlop (1983) defines the parameter as $(MDF_{NRM} - MDF_{IRM})/MDF_{NRM}$. BM behavior represents samples containing both SD and MD magnetic carriers. Dunlop (1983) shows that this parameter is negative for MD, between 0 and 0.2 in the case of SD or mixed behavior, and larger than 0.3 in the case of BM.

Scanning electron microscope (SEM) analysis using a JEOL JSM-5900LV instrument at the Geological Survey of Finland was carried out to further constrain the nature of the magnetic carriers.

Chemical analyses of the samples were done by standard X-ray fluorescence spectrometry at the School of Geosciences, University of Witwatersrand, Johannesburg. Details of the analytical method and accuracies and precisions can be found in Reimold and Colliston (1994).

4. Petrophysical results

Petrophysical properties of the various lithologies are compiled in Table 1. The most pronounced observation is the high intensity of NRM values (Fig. 3a), which is observed in several lithologies independently of the distances from the center of the structure (Fig. 4a) and in both the Vredefort and the Johannesburg domes. As the Koenigsberger's ratio (Q value) is the ratio between NRM and induced magnetization (susceptibility), these results lead to high Q values in these rocks (Figs. 3a and 4a). The high Q values are due to high remanences as the susceptibilities are typical for these lithologies. The high variation of Q values with the distance from the impact center suggests that the high NRM intensity is not directly linked to the impact event.

4.1. Impactites

Vredefort Granophyre (impact melt rock) samples show one order of magnitude higher intensity of NRM than PT breccia samples, but their susceptibilities are of the same order of magnitude

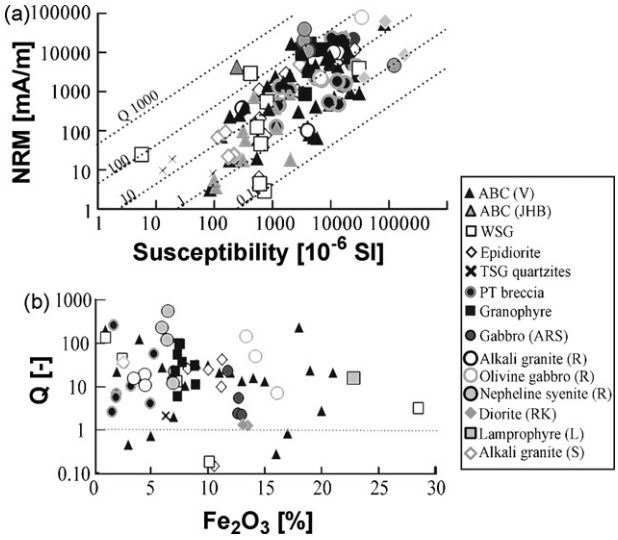


Fig. 3. (a) Natural remanent magnetization vs. susceptibility of the various lithologies of the Vredefort and Johannesburg domes. Q values are drawn as dotted lines. (b) Q value as a function of iron content. The dotted horizontal line marks the typical Q value for granites. ABC, Archean Basement Complex; V, Vredefort; JHB, Johannesburg Dome; WSG, Witwatersrand Supergroup; TSG, Traansvaal Supergroup; ARS, Anna's Rust Sheet; R, Rietfontein; RK, Roodekraal; S, Schurwedraii; L, Lindeques Drift.

(Fig. 3a). Similar observations regarding susceptibilities were made by Henkel and Reimold (1998, 2002). This results in higher Q values for the Granophyre (~10–15). This observation might be in part explained by the facts that the iron content (Fe_2O_3) of the Granophyre samples (on average: 7.7 wt%) is larger than that of PT breccia samples (on average: 2.9 wt%) (Fig. 3b), or that the Granophyre is melt that has acquired thermoremanent magnetization, whereas magnetization of PT breccias may have scattered directions.

4.2. Other samples from the Vredefort and Johannesburg domes

Archean basement granites and gneisses of the Vredefort Dome show a wide range of Q values, from 0.3 to 235 (average: 35), whereas granites usually have Q values below 1 (e.g., Clark and Emerson, 1991; Puranen, 1989). For comparison with the Vredefort Dome samples, similar Archean basement rocks (gneisses and migmatites) from the Johannesburg Dome were analyzed in this study. Surprisingly, these rocks also show elevated Q values, the average value being 46, and the range extending from 0.1 to 434 (Table 1 and Fig. 3a). Similarly, the majority of the samples from WSG, TSG, VSG lithologies, and all intruded bodies (except Roodekraal) show high Q values (Fig. 3a and Table 1). This peculiar petrophysical nature of the basement rocks of the Vredefort structure has been reported in a number of earlier studies (e.g., Carporzen et al., 2005, 2006; Hargraves, 1970; Hart et al., 1995, 2000; Jackson, 1982), however, it has never been recognized as a regionally more widespread phenomenon before this study.

No linear correlation results when the Q value is plotted against the iron content (see Fig. 3b). Moreover, magnetic hysteresis data show that there is no correlation between high Q value and domain state (Fig. 4). Hysteresis ratios (M_{rs}/M_s and H_{cr}/H_c) of studied samples do not show dependence on the distance from the approximate center point of the structure (Fig. 4a and b) and, consequently, no dependence on the elevated temperatures and shock pressures caused by the impact event.

Table 1
Petrophysical and rock magnetic properties of the pre-impact rocks, igneous intrusives, the Vredefort impactites and post-impact intrusive rocks of Vredefort and Johannesburg domes.

D (kg m^{-3})	k ($\times 10^{-6}$ SI)	J_r (mA/m)	Q	N/n	M_{rs}/M_s	H_{cr}/H_c	n
Host rocks							
ABC (V) granite, gneisses							
2628	6373	2763	32	33*/156	0.13	4.00	13
2568–2724	85–30,866	3–16,915	0.3–235		0.01–0.35	2.65–6.17	
ABC (JHB) granite, gneisses							
2669	651	568	46	11*/48	0.17	3.85	7
2605–2812	100–2027	0.6–4255	0.1–434		0.01–0.62	1.15–7.37	
LWSG metapelite, quartzite							
2737	3807	839	38	9*/42	0.44	1.51	3
2595–2838	−7 to 30,548	2.9–3891	0–175		0.2–1.0	1.11–1.89	
UWSG epidiorite							
2978	6859	28,554	46	13*/58	0.28	2.54	5
2863–3059	580–26,516	4–319,952	0.2–294		0.09–0.59	2.34–2.87	
WSG (JHB) shale							
3298	124,651	58,179	15	3*/6	0.27	1.88	2
3062–3648	61,071–227,572	29,341–73,918	8–30		0.16–0.38	0.61–3.15	
TSG quartzite							
2589	36	7	8	4*/13	0.11	1.35	2
2558–2644	−0.3 to 94	2–9	2.1–18.5		0.0003–0.22	1.07–1.64	
VSG (V + JHB)							
2921	557	14	1	4*/18			
2878–2955	534–605	0.5–27	0.02–1.28				
Pre-impact intrusions							
Shurwedraai alkali granite							
2600	721	1039	15	5*/24			
2568–2620	116–2940	22–4990	2.8–43				
Rietfontein olivine gabbro							
3074	23,821	96,564	79	3*/23			
2939–3161	6727–34,006	2206–208,464	8–171				
Rietfontein alkali granite							
2627	12,559	8909	18	3*/35			
2594–2645	11,125–13,410	6746–10,097	12–22				
Lindeques Drift mafic intrusion							
3216	135,743	35,634	10	2*/7			
3163–3270	85,883–185,603	8602–62,667	1–18				
Roodekraal diorite							
2832	38,467	2290	1	2*/2			
Impactites							
Pseudotachylitic breccia							
2654	8458	4131	29	15*/81	0.12	4.89	4
2626–2737	300–27,640	125–23,376	0.9–259		0.04–0.16	3.04–8.65	
Granophyre							
2800	8198	11,080	43	11*/49	0.06	4.23	6
2784–2839	3197–16,471	860–8061	6.0–96		0.04–0.07	3.58–5.35	
Post-impact intrusive							
ARS gabbro							
2999	20,796	28,007	30	5*/34	0.27	1.84	3
2974–3015	15,074–24,537	1402–129,025	2–139		0.14–0.40	1.59–2.07	
Rietfontein nepheline syenite							
2627	12,559	14,971	231	5*/32			
2598–2611	4014–3545	10,861–39,082	66–273				

Note: The first row indicate the mean value for the rock type and the second row show minimum and maximum values of petrophysical property. D, density; k, susceptibility; J_r, intensity of the natural remanent magnetization; Q, Koenigsberger's ratio calculated for a field of 50 μT; N/n, samples/specimens; *, the level used in statistical calculations; M_{rs}, saturation remanence; M_s, saturation magnetization; H_{cr}, coercivity of remanence; H_c, coercive force. Number of specimens in rock magnetic study was lower than in petrophysical studies. ABC, Archean Basement Complex; V, Vredefort Dome; JHB, Johannesburg Dome; (L/U)WSG, (Lower/Upper) Witwatersrand Supergroup; TSG, Traansvaal supergroup; VSG, Ventersdorp supergroup; ARS, Anna's Rust Sheet.

To investigate further whether the high Q values are solely associated with single domain behavior we examined the correlation between Q values and hysteresis parameters (Fig. 4e). Interestingly, samples with very high Q values plot mainly in the pseudo-single-domain (PSD) region in the vicinity of the SD–MD mixing line, in agreement with other observations indicating the existence of two grain sizes of magnetite. In contrast, samples with "normal" Q values are scattered on the Day plot.

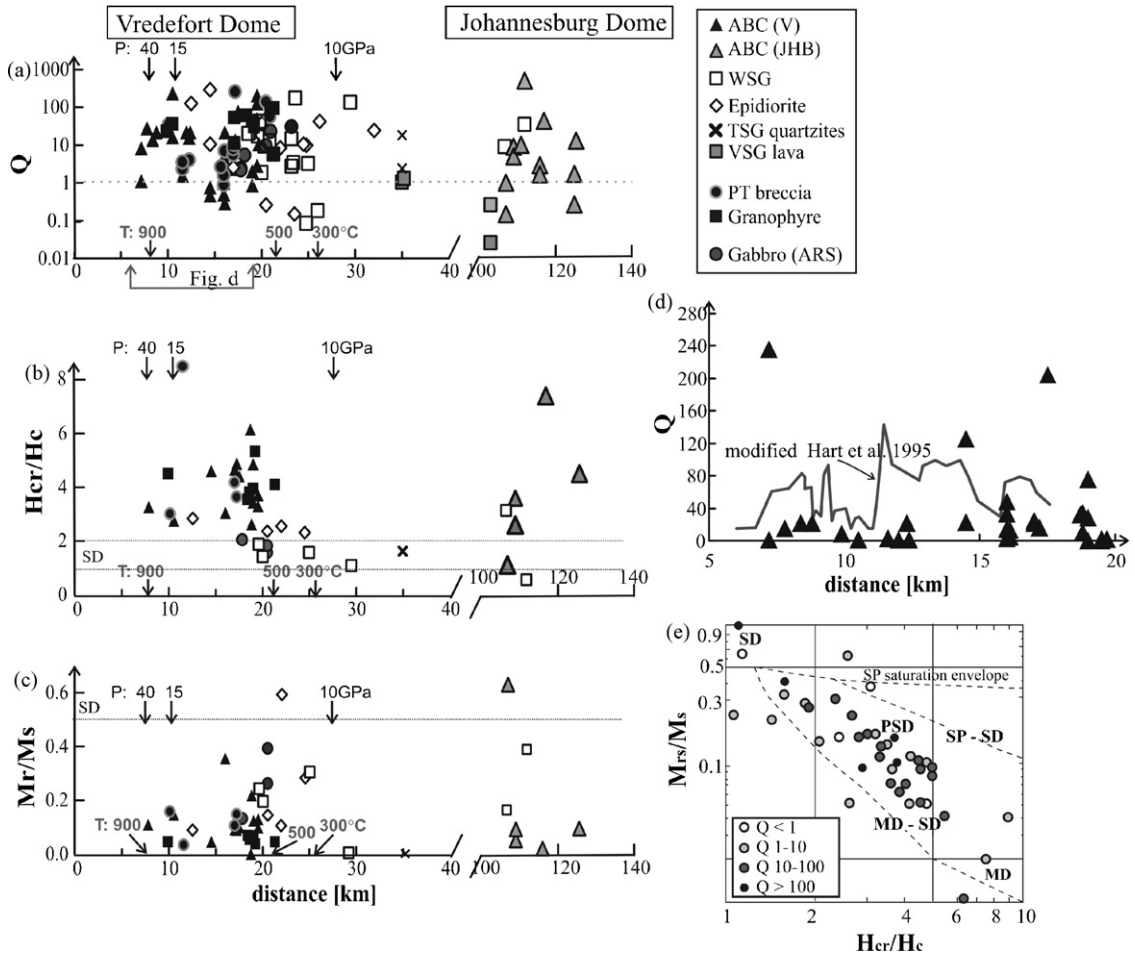


Fig. 4. Rock magnetic properties. (a) Q value vs. distance, (b) H_{cr}/H_c vs. distance, (c) M_{rs}/M_s vs. distance, (d) Q values of ABC samples vs. distance from this study and study by Hart et al. (1995), and (e) M_{rs}/M_s and H_{cr}/H_c for the samples from the Vredefort and Johannesburg domes plotted on the Day-plot (Day et al., 1977; Dunlop, 2002) indicating the different Q values. Upper black arrows mark the pressure zones and lower black arrows mark the temperature zones within the structure (modified after Gibson and Reimold, 2005). Dotted horizontal line marks typical Q value for granites. Grey lines in (b) and (c) mark typical hysteresis values for single domain (SD) magnetite. Note that the scale in x-axis in (a)–(c) changes at 40 km. M_{rs}, saturation remanent magnetization; M_s, saturation magnetization; H_{cr}, coercivity of remanence; H_c, coercive force; P, pressure; T, temperature; ABC, Archean Basement Complex; V, Vredefort; JHB, Johannesburg Dome; WSG, Witwatersrand Supergroup; TSG, Traansvaal Supergroup; VSG, Ventersdorp lavas; ARS, Anna's Rust Sheet.

5. Rock magnetic results

5.1. Thermomagnetic analyses

High temperature thermomagnetic analyses were carried out for all lithologies of the Vredefort and the Johannesburg domes. Low temperature thermal treatment was carried out for selected ABC and impact generated rock samples. Samples were selected so that they included cases with high and with normal Q values. The behavior of the investigated samples is very variable. Magnetite is present in most samples, however the mineralogical stability varies in accordance with the presence of other magnetic phases such as pyrrhotite.

5.1.1. Impactites

Representative results of thermomagnetic analyses (susceptibility vs. temperature) of PT breccia and Granophyre samples are shown in Fig. 5. PT breccia samples show mainly reversible heating and cooling curves (as exemplified by sample LV44, Fig. 5a). They show Hopkinson peaks indicating single domain (titano)magnetite with Curie points between 550 and 570 °C as magnetic carrier phase. Two PT breccia samples with normal Q values from Salvamento quarry (e.g., V145, Fig. 5b) show kinks around 130–170 °C, which is a typical low-temperature titanomagnetite (maghemitization) phenomenon (Ade-Hall et al., 1971; see also Letts et al., 2005), possibly related to hydrothermal alteration. These samples show also a decrease of susceptibility between 320 and 350 °C, which is probably related to the inversion of maghemite (e.g., V145, Fig. 5b). Low-temperature measurements of PT breccia samples show the Verwey transition at temperatures between 100 and 105 K (Fig. 5a and b). Interestingly, sample V145 from Salvamento quarry (Q value: 2.6) shows two Verwey transitions: one at 100–103 K and the other at 122–125 K (Fig. 5b), indicating that there are two populations of magnetite grains with different grain sizes and/or lattice structure (e.g., Dunlop and Özdemir, 1997).

Granophyre samples show reversible heating and cooling curves, Hopkinson peaks, and Curie points between 560 and 580 °C, indicating single domain (titano)magnetite (e.g., sample LV116, Fig. 5c) as the prominent phase. These samples yielded more homogeneous demagnetization curves than the PT breccia samples, and reversible heating and cooling curves. Low-temperature measurements revealed the Verwey transition between 100 and 105 K (Fig. 5c), indicating also magnetite.

5.1.2. Archean gneiss samples from the Vredefort Dome

Some ABC samples from the Vredefort Dome show strong irreversibility of heating and cooling curves indicating that mineralogical changes took place during the heating experiments. Many samples show a slight increase of susceptibility around 120–170 °C (e.g., samples LV4 "biotite granite, $Q = 2$" and LV92 "granite gneiss, $Q = 11$"; Fig. 6a and b), which is a typical low-temperature titanomagnetite (maghemitization) phenomenon (Ade-Hall et al., 1971). These samples also show a decrease of susceptibility around 320–350 °C, which may be related to the inversion of maghemite. Intensity of susceptibility of the cooling curve is usually higher in such samples, indicating that new magnetite formed during heating. On the other hand, quite a few ABC samples show reversible heating and cooling curves indicating that no mineralogical changes took place during the treatments (e.g., LV46 "charnockitic gneiss, $Q = 2.9$"; Fig. 6c).

Low-temperature treatments show that there are samples characterized by two different behaviors during the experiment. Two of the studied samples (LV4 and LV92) show two distinct Verwey transitions at temperatures of 100–102 K and 122–125 K, respectively (Fig. 6a and b). However, the majority of the samples shows only one Verwey transition either around 110–113 K (e.g., LV46, Fig. 6c) or around 100–102 K. Carporzen et al. (2006) noted two Verwey transitions in the majority of their ABC samples. In this study it was the case only for 2 of the 12 studied ABC samples.

Following the idea of Carporzen et al. (2006) we repeated the low temperature susceptibility treatment on some samples that had been subjected to the high temperature experiment, in order to investigate the stability of the two magnetic phases. Our results

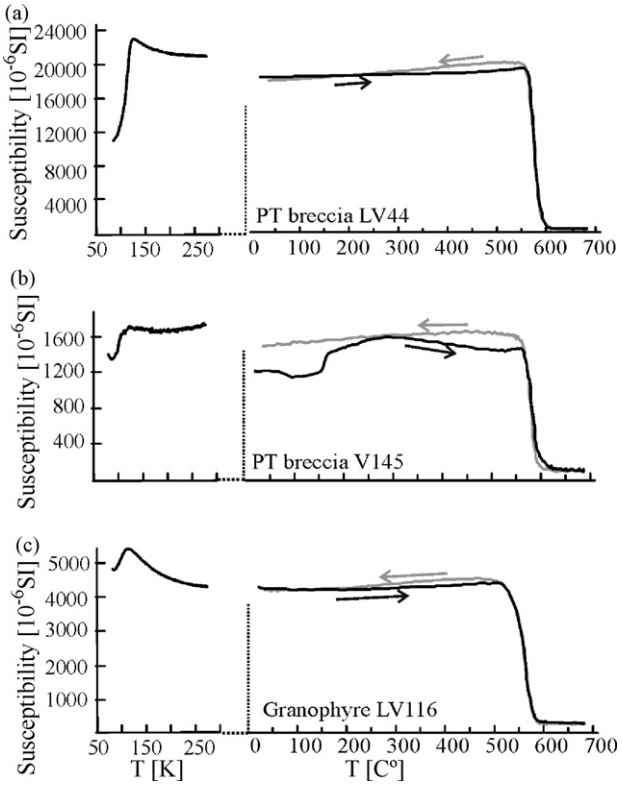


Fig. 5. Thermomagnetic curves (susceptibility vs. temperature) for impact lithologies of the Vredefort structure. Dotted horizontal and vertical lines indicate the change in the scale of the *x*-axis. In the first part of the experiment specimens were heated from the temperature of the liquid nitrogen up to 0 °C. In the second part specimens were heated from room's temperature up to 700 °C and cooled back to room's temperature. (a) PT breccia sample LV44, (b) PT breccia sample V145 and (c) sample LV116.

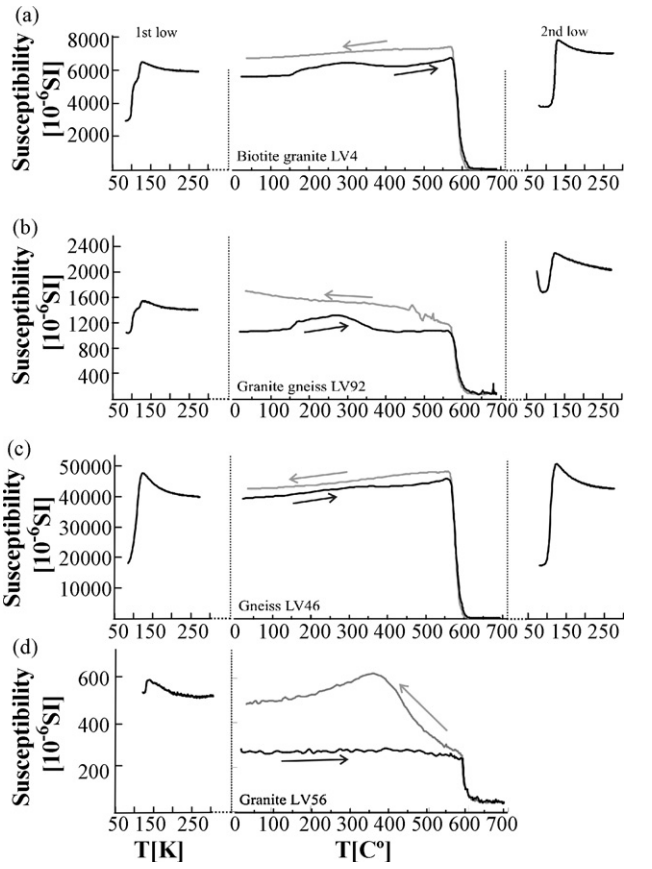


Fig. 6. Thermomagnetic curves (susceptibility vs. temperature) for ABC samples of Vredefort structure. Dotted horizontal and vertical lines indicate the change in the scale of the *x*-axis. In the first part of the experiment specimens were heated from the temperature of the liquid nitrogen up to 0 °C. In the second part specimens were heated from room temperature up to 700 °C and cooled back to room temperature. In the third part specimens were again heated from the temperature of liquid nitrogen up to 0 °C. (a) Biotite granite sample LV4, (b) granite gneiss sample LV92, (c) gneiss sample LV46 and (d) gneiss sample LV56.

indicate that samples showing two Verwey transitions in the first low-temperature treatment display only one Verwey transition during the second low-temperature treatment (Fig. 6a and b). The disappearance of the second phase is probably a result of mineralogical alteration. The lower temperature transition is removed and the higher temperature transition is now seen at a few degrees lower than in the first low-temperature treatment (at 120–122 K). Some samples that showed a Verwey transition around 100–102 K in the first low-temperature treatment show a Verwey transition around 122 K as a result of the second low-temperature treatment. For comparison, this experiment was also carried out on some specimens showing only one Verwey transition. Fig. 6c shows that this transition is stable against heating.

All the studied samples also show somewhat elevated susceptibility values during the second low-temperature treatment compared to the measurements during the first, which is most probably due to the physico-chemical changes during the high-temperature treatment.

5.1.3. Archean basement samples from the Johannesburg Dome

The ABC samples from the Johannesburg Dome showed variable behavior during the thermal experiment. Some samples show strong irreversibility of heating and cooling curves. Intensity during cooling in such samples is much higher than intensity during heating, indicating that magnetite forms during heating (e.g., sam-

ple LV56 "granite, $Q = 39$"; Fig. 6d). However, a few samples showed reversible heating and cooling curves, with a distinct Hopkinson peak indicating that no mineralogical changes took place during the treatment and that single domain magnetite is present in the samples. Only two (e.g., sample LV56; Fig. 6d) of the four studied samples showed a Verwey transition at ca. 120 K during the low-temperature treatment. This indicates that the structure and lattice of magnetite is somewhat distorted in these samples (e.g., Dunlop and Özdemir, 1997).

5.1.4. Other studied samples

Two epidiorite samples, as well as quartzite samples from the TSG and from the WSG showed irreversible behavior, whereas one epidiorite sample (LV83) with high Q value (127) showed a Hopkinson peak and a Curie temperature of 570 °C and nearly reversibility during the experiment. In contrast, WSG shales showed nice reversible behavior, a Hopkinson peak, and a Curie temperature of 580 °C indicating single domain magnetite. Shale shows also the additional presence of a pronounced peak of susceptibility at about 305 °C, which may indicate a pyrrhotite component.

The ARS gabbro samples showed irreversible heating and cooling curves, indicating mineral alterations during heating. The heating curves showed a decrease in susceptibility between 320 and 330 °C and may indicate the presence of higher Ti-content in titanomagnetite or the presence of pyrrhotite. Another phase was observed between 560 and 575 °C, which is indicative of (titano)magnetite. Pybus (1995) and Reimold et al. (2000) described the presence of titanomagnetite and ilmenite in ARS samples. Our thermal paleomagnetic experiments suggest also the presence of hematite.

5.2. Other rock magnetic studies

Using SEM we analysed the magnetic mineralogy of four PT breccia, two Vredefort Granophyre, and altogether eight ABC samples from both the Vredefort and Johannesburg domes, as well as two ARS gabbro specimens. The study verified the presence of titanomagnetites, with variable Ti-content and variable grain sizes in all specimens. This was further proven by other rock magnetic studies (hysteresis analyses, Lowrie–Fuller test and MDF-parameter calculations). In general, the shapes of the hysteresis loops of studied samples are coherent with magnetite as a dominant phase, as they do not show a wasp-waisted behavior, which would be the case for the presence of multiple magnetic carriers (e.g., Dunlop and Özdemir, 1997). However, some ABC samples showed typical MDF-parameter values (0.3–0.5) for BM behavior, indicating that these samples contain both SD and MD grains of magnetite. Such samples have comparatively lower Q values from 0.7 to 14. In the case of the studied ABC samples, the IRM is harder than the ARM. This supports the observation made above that magnetic carriers are mainly MD size or both MD and SD size.

The decay curves of intensity of NRM and for both the ARM and IRM decay were examined. It was expected that the NRM decay curve of samples with high Q values would be similar to the decay curve of IRM, as the high remanent magnetization values resulting in high Q values were expected to be caused either by lightning or by other high fields, such as an impact-generated plasma field proposed by Carporzen et al. (2005). Among the samples showing only one population of magnetic carriers this was clearly the case only for one granite specimen. The NRM intensity of other specimens decayed more as ARM than IRM, indicating that the magnetization is of thermal origin.

6. Paleomagnetic results

6.1. Impact lithologies

Six PT breccia samples out of 16 and 8 Granophyre samples out of 14 were excluded from paleomagnetic analysis, as they yielded high Q values between 20 and 259 and anomalous directions of magnetization. The rest of the impact-generated rocks of the Vredefort Dome gave stable and congruent paleomagnetic results (Table 2, and Figs. 7 and 8). A high coercivity (50–100/160 mT) and high-temperature (up to 520–580 °C) component ($D = 18.3°$, $I = 54.8°$, $\alpha_{95} = 8.1°$) yields a paleomagnetic pole of Plat = 25.1° and Plong = 43.5° ($A_{95} = 10.6°$) for results from 16 samples.

PT breccia samples behave similarly during both alternating field (AF) and thermal (TH) demagnetization, showing two components of magnetization. The first one, a low coercivity and low temperature component, is most likely of viscous origin in most of the samples and shows random directions. The second, high-coercivity and high-temperature component is stable and shows well-defined NNE declination and positive, moderately steep inclination (Figs. 7a and b and 8). Thermal treatment shows that this remanence resides in magnetite (Fig. 7b).

Granophyre samples also show two components during both AF and TH demagnetization. The high-coercivity and high-temperature component is similar to that obtained for PT breccias and resides in magnetite (Figs. 7c and 8). However, the samples show a low-to-intermediate coercivity component slightly harder than the one seen in PT breccias. The direction of this low to intermediate component of Granophyre samples is variable and in the majority of the samples it is interpreted to be caused by viscous magnetization.

The direction of the observed low coercivity component varied substantially between samples, some showing low and others showing high inclinations of both reversed and normal polarity. Directions were highly scattered. However, it is worth noting that a slight trend was observed in these low coercivity components; a few samples – characterized by both normal and high Q values – showed shallow northerly (for example sample LV16, see Fig. 11a) or shallow southerly directions. A similar component was also seen in some target rock samples of the Vredefort Dome and will be discussed further below.

6.2. ABC samples

6.2.1. Impact-related direction in ABC samples

The impact-related (or Bushveld-related) direction as found in the impact-produced rocks was separated from some of the ABC samples. We prepared several specimens out of one hand sample and accepted the direction to be impact-related if all the measured specimens from such a single sample showed congruent directions. Due to the fact that many of the rocks from the ABC show anomalously high Q values, we were only able to reliably separate this component from four samples showing reliable NRM intensity values. An example of the behavior of granite pegmatite clasts in PT breccia (sample LV40) during demagnetization is shown in Fig. 9a. This sample was taken 16 km from the assumed center of the structure (compare Fig. 1). The same direction was also seen in some ABC samples taken from the Vredefort Dome (for example in sample LV12; Fig. 9b, 14.5 km from the center of the structure). The mean direction of this component in the ABC samples is $D = 2.1°$, $I = 62.5°$ ($\alpha_{95} = 7.2°$; 4 samples and 14 specimens), where it is isolated at 50–100/160 mT cleaning range. This direction is very close to the direction isolated from the impactites. We consider this as proof that the temperature in the presently exposed rocks rose after the impact sufficiently high to thermally or thermochemically remag-

Table 2
Palaeomagnetic results for the Vredefort impactites and impact related component of magnetization in Archean basement samples.

Sample	Rocktype	Lat (°)	Long (°)	*n*	*D* (°)	*I* (°)	*k*	α_{95} (°)	Plat (°N)	Plon (°E)	A_{95} (°)
Impactites											
LV116	Granophyre	26°50.3	27°27.3	2	60	48			9	78	
LV126	Granophyre	27°00.9	27°22.7	2	356	60			22	24	
LV63	Granophyre	26°53.9	27°21.8	5	23	53	805	3	26	49	3
LV64	Granophyre	26°53.9	27°21.8	5	26	44	11	24	31	55	23
V142	Granophyre	26°50.9	27°31.8	4	13	60	47	13	21	38	18
V143	Granophyre	26°50.9	27°31.8	4	28	64	581	4	13	47	5
LV11	PT breccia	26°53.9	27°24.2	6	22	53	137	6	26	48	7
LV18	PT breccia	26°56.7	27°25.5	5	12	51	23	16	31	39	18
LV35	PT breccia	26°53.4	27°24.0	4	9	58	187	7	24	35	9
LV38	PT breccia	26°53.9	27°24.5	4	12	63	77	11	18	37	15
LV43	PT breccia	26°53.9	27°24.5	3	32	54	28	24	22	55	28
LV44	PT breccia	26°58.9	27°22.1	2	348	55			27	17	
LV45	PT breccia	26°58.9	27°22.1	4	27	59	285	5	19	50	7
V145	PT breccia	26°53.4	27°24.1	3	351	60	2361	3	22	20	3
OT1	PT breccia			2	31	44			30	59	
V144	PT breccia	26°50.9	27°31.8	1	37	47			25	63	
Mean				56	18.3	54.8	30.0	8.1	25.4	43.8	9.6
ABC-secondary											
LV12	Gneiss	26°54.6	27°25.5	8	6	66	56	7	15	31	11
LV39	Gneiss	26°53.9	27°24.5	1	12	65					
LV40	Granite	26°53.9	27°24.5	4	5	62	249	6	20	32	8
LV41	Granite	26°53.9	27°24.5	1	349	56					
Mean				14	2.1	62.5	163.5	7.2	19.2	28.9	9.9

Lat and Long, latitude and longitude of the sampling site; *n*, number of specimens in case of same sampling site results have been combined before calculating the mean; *D* and *I*, declination and inclination of the remanent magnetization; *k*, Fisher's (1953) precision parameter; α_{95}, radius of circle of 95% confidence of direction; Plat and Plong, the latitude and longitude for the paleomagnetic pole; A_{95}, radius of the circle of 95% confidence of the pole. ABC, Archean Basement Complex.

netize these rocks, which is in disagreement with the suggestion of Carporzen et al. (2006) and model by Muundjua et al. (2007).

6.2.2. Other directions of magnetization in ABC samples

Archean basement samples from the Vredefort and Johannesburg domes, with high Q values show variable directions of hard coercivity magnetization (Fig. 10a). This has been reported also for earlier paleomagnetic studies (Carporzen et al., 2005; Hart et al., 1995, 2000; Hattingh, 1989, 1999; Jackson, 1982; Layer et al., 1988, 1989a,b). However, the majority of our specimens show congruent directions of this high coercivity magnetization within one sample, which is in contrast to the observation of Carporzen et al. (2005) and Hart et al. (2000), who reported variable directions at the centimeter scale in all their samples.

In addition to this hard coercivity component our samples also show a low coercivity component. Such a component has variable directions with mainly low and steep inclinations of normal and reversed polarity. Despite these highly scattered NRM directions we were able to see a slight trend of directions of magnetization for some samples from both the Vredefort (samples LV4, LV5, LV10, LV48, LV84, LV93 and LV94) and Johannesburg (samples LV56, LV58 and LV59) domes towards either a shallow southerly upward or a

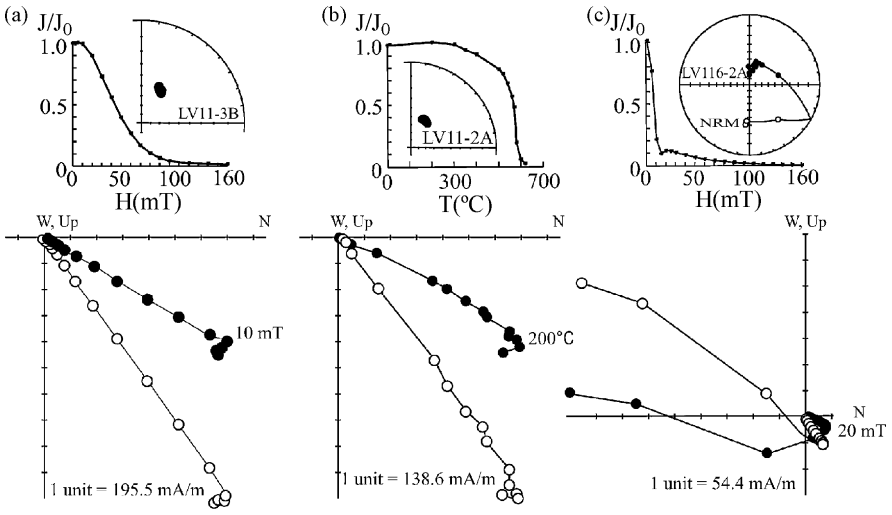


Fig. 7. Examples of demagnetization behavior of pseudotachylitic breccia and Granophyre specimens. Shown are a stereoplot, intensity decay vs. used alternating field (AF)/temperature, and orthogonal (Zijderveld, 1964) demagnetization diagrams, where open (closed) symbol denotes vertical (horizontal) planes. (a) Specimen LV11-3B (Q = 8.5) represents the typical behavior of the pseudotachylitic breccia specimens during AF treatment, (b) specimen LV11-2A (Q = 6.8) represents typical behavior of the pseudotachylitic breccia specimens during thermal treatment and (c) Granophyre specimen LV116-2A (Q = 6.0) represents typical behavior of granophyre during AF treatment. Number at demagnetization step denotes the AF value or temperature.

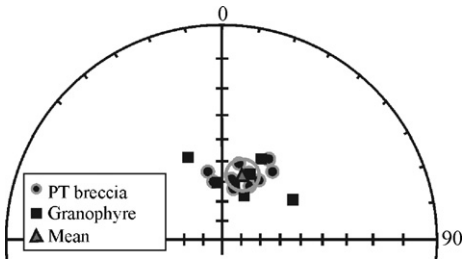


Fig. 8. Site mean directions of the impactite-type component from pseudotachylitic (PT) breccia and Granophyre samples from the Vredefort Dome (see Table 2). Closed symbol denotes downward direction of magnetization. Triangle denotes mean of all and circle around it is circle of 95% confidence of direction. Equal angle projection.

shallow northerly downward (reversed to the first one) direction. Specifically such a direction was seen in the samples collected close to the Anna's Rust Sheet gabbro intrusion, which leads us to postulate that this effect is caused by heating from the ARS. In some samples such a direction was seen as a hard coercivity component (for example gneiss sample LV48-1B, see Fig. 11b), whereas other samples showed such a direction as an intermediate (for example in sample LV56-1A, see Fig. 11c) or a low coercivity component. A similar trend was also observed in some of the Vredefort Granophyre (Fig. 11a) and PT breccia samples. Samples showing this component may have either normal or elevated Q values (up to 40). However, they do not correspond to the highest Q values observed in the area. The combined mean direction ($D = 160^\circ$, $I = -5^\circ$, $\alpha_{95} = 21^\circ$) of target and impact rock samples was calculated.

6.3. Other pre-impact samples

Of 14 epidiorite samples 11 showed high Q values (from 8.5 up to 294), and only three samples maintained reasonable Q val-

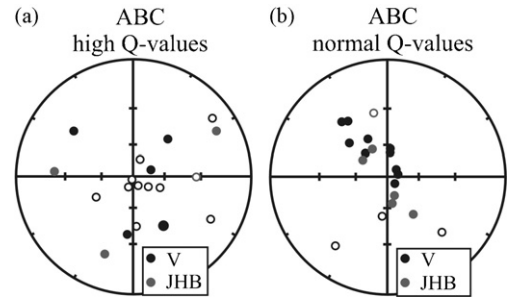


Fig. 10. Mean directions of the samples from the Archean Basement Complex (ABC) of the Vredefort (V) and Johannesburg (JHB) domes. (a) Magnetization direction of samples with high values of NRM intensity and (b) magnetization direction of samples with normal values of NRM intensity. Closed (open) symbol denotes downward (upward) direction of magnetization.

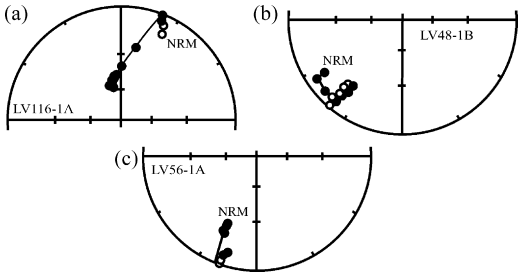


Fig. 11. Shallow direction interpreted as an 1.1 Ga overprint. (a) Vredefort Granophyre sample LV116-1A ($Q = 5.8$) showing a shallow northerly downward low coercivity component, (b) gneiss sample LV48-1B ($Q = 1.1$) from the Vredefort Dome showing a shallow southerly upward hard coercivity component, and (c) granite sample LV56-1A ($Q = 34$) showing shallow southerly upward intermediate/hard coercivity component.

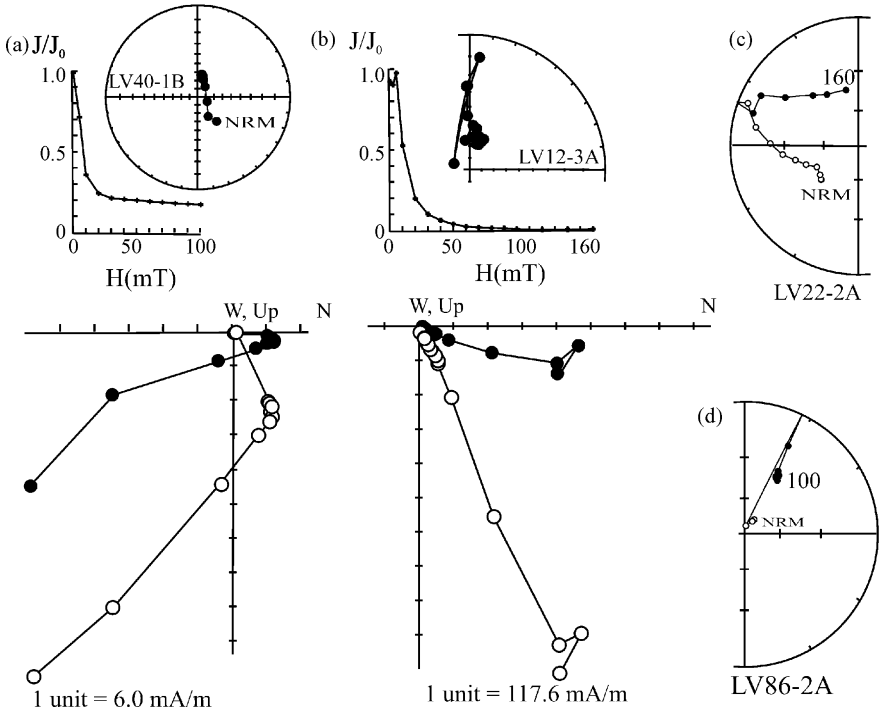


Fig. 9. Examples of demagnetization behavior of several lithologies showing the impactite-type component during the alternating field treatment. In (a) and (b) a stereoplot, intensity decay vs. used alternating field (AF)/temperature, and orthogonal (Zijderveld, 1964) demagnetization diagrams are shown, where open (closed) symbols denote vertical (horizontal) planes. In (c) and (d) shown are only stereoplots. (a) Specimen LV40-1B is a granite pegmatitic clast ($Q = 2.6$) sampled from a meter wide PT breccia occurrence, (b) specimen LV12-3A is a biotite-rich gneiss ($Q = 1.0$), (c) specimen LV22-2A is a grey gneiss ($Q = 13.0$) and (d) specimen LV86-2A is a quartzite ($Q = 3.1$) of the Turfontein Subgroup in the Witwatersrand Supergroup.

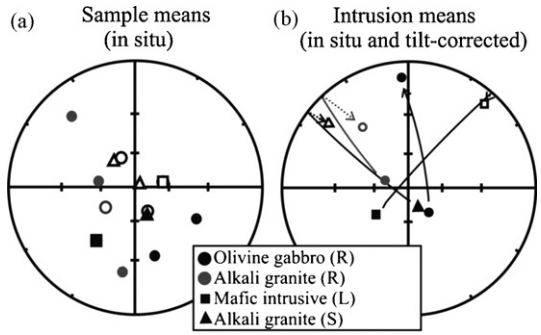


Fig. 12. (a) In situ mean directions of the samples from the Rietfontein (R, alkali granites and olivine gabbros), Lindeques Drift (L, mafic intrusive) and Schurwedraai (S, alkali granites) complexes, and (b) in situ and tilt corrected mean values for the intrusions. Used strike/dip for Rietfontein is 75°/overturned dip 70°SE (Hargraves, 1970), Lindequesdrift is 65°/overturned 45°SW (Hargraves, 1987) and Schurwedraai 45°/overturned 60°SE. Closed (open) symbol denotes downward (upward) direction of magnetization.

ues of 0.3–2.6. These three samples did not produce any stable paleomagentic results. Other samples from the WSG ($n = 11$), TSG ($n = 4$), and VSG ($n = 4$) behaved in complex ways, as they all showed high Q values (Fig. 3a). The number of samples producing useful results is so small that we cannot draw any further reliable conclusion.

6.4. Anna's Rust Sheet

We studied six samples from the ARS gabbro. Three of them have Q values between 10 and 140. In spite of these anomalously high Q values, five ARS samples yielded a reasonably clustered direction of magnetization $D = 330.7°$, $I = 63.1°$ ($\alpha_{95} = 17.8°$). Specimens from one sample also show congruent directions. The ARS has been dated at 1108.6 ± 1.2 Ma (Hanson et al., 2004b), which leads us to expect that the magnetization direction would be close to that known for the 1.1 Ga Umkondo igneous activity (the shallow northerly or southerly Umkondo-type direction; Hanson et al., 2004b), but this is not the case. It is most probably caused by some later remagnetization events, but not related to the known, ca. 185 Ma Karoo volcanism ($D = 156°$ and $I = 53°$; Hargraves et al., 1997). Magnetization of these samples is rather soft showing only a single component, indicating that its origin may be lightning. More promising results of ca. 1.1 Ga overprint were obtained from some of the ABC rocks as was pointed out earlier (see Section 6.2).

6.5. Alkaline and dioritic intrusions

In situ directions of the characteristic remanent magnetization (ChRM) of the studied samples are shown in Fig. 12a. Mean values for the rock types are shown in Fig. 12b. ChRM directions of the samples from the Rietfontein complex, from the Lindeques Drift mafic intrusion, and from the Schurwedraai alkali granite complex show wide between-sample scatter (Fig. 12a) and high Q values (from 8 to 660), which indicate that the remanence of such samples may have been reset. Two dioritic samples with normal Q value (about 1.5) from the Roodekraal Complex show also wide between-sample scatter, which has been reported also by Hargraves (1970). However, because we see high Q values and as the sample number is so small, we cannot draw any further firm conclusion from these results.

7. Discussion

7.1. High Q values and origin of magnetization

It is obvious that values of NRM and, hence, Q values for many samples of the pre-, syn- and post-impact lithologies of the Vredefort structure – and as determined for the first time, of the Johannesburg Dome 120 km to the northeast of it – are elevated compared to the values for similar rock types found elsewhere (see Table 1 and Figs. 3a and 4a). High values of NRM intensity and random directions of magnetization of ABC samples from the Vredefort Dome have already been noted in previous studies (e.g., Carporzen et al., 2005; Hart et al., 1995, 2000; see Fig. 4d for comparison). According to Carporzen et al. (2006) lightning could explain about 26% of the observed high NRM values. They obtained this number by defining the number of samples from impact lithologies showing high NRM values. They concluded that the remaining 74%, which represent the host rocks with high Q values, were affected by an impact-generated high magnetic field.

Cloete et al. (1999) and Hart et al. (1995, 2000) suggested that ultra-small single-domain magnetite formed in a high pressure/temperature environment and crystallized along planar deformation features (PDFs) is the most likely source of the high remanence in the Vredefort Dome's ABC samples. Hart et al. (1995) presented a profile showing connection between high Q values and small grain sizes. They also postulated that these values depended on the distance from the center of the structure. Our results contradict these findings (Fig. 4): high Q values are independent of both the hysteresis ratios indicating small grain sizes and of distance from the center of the structure (Fig. 4). In addition to Hart et al. (2000), Carporzen et al. (2005) suggested that although PDFs control the distribution of magnetite in shocked rocks, a plasma field produced from the impact event generated small-wavelength magnetic fields of high intensity, whose direction varied over centimeter scales. This field randomized the magnetic remanence of the magnetite grains in PDFs within cm-scale regimes, as shown from the high within-sample scatter that they observed. In contrast, about 80% of our ABC samples show good within-sample agreement, but between-sample directions scatter among the samples independently of their Q value.

As, firstly, we could not observe a concentration of high Q values near the center of the structure, as should be the case if impact induced these high Q values, and, secondly, the high Q values are now also seen in samples from the Johannesburg Dome (Fig. 4), the direct Vredefort shock connection (unless the Johannesburg Dome would also represent an impact structure, etc.) may be ruled out. This is also supported by the fact that there is no correlation between hysteresis data and high Q values (Fig. 4) of the basement rocks, as would be expected if the ultra-fine particles in the PDFs were the carriers of the high Q values. Moreover, the elevated NRM values of impactites cannot be explained by the plasma-induced magnetic field, as they were covered since impact times by 5–10 km of rock and cooled much slower (several hundreds of thousands of years; e.g., Ivanov, 2005) than the life time of a plasma field (from seconds to minutes; e.g., Crawford and Schultz, 1999; Turtle and Pierazzo, 1998). Another explanation for high Q values could be lightning, but we discount it as the cause of all observed high Q values (expect in the case of ARS samples and some of the ABC samples) based on the Lowrie–Fuller test and the fact that all the studied lithologies from all the surface exposures, as well as some samples (ABC and PT breccia) from the quarries show the elevated Q values. However, quarries did not show as high Q values ($4 < Q$ value < 50) as samples from the surface. Moreover, studied samples showed relatively hard magnetization with multiple remanent magnetization components, which is in contrast to lightning-induced magnetiza-

tion (soft and normally single component; e.g., Irving, 1964). As we exclude lightning as a cause for all the observed high Q values, a possible explanation for the high Q values could be elevated temperatures and possible hydrothermal fluid circulations within the two uplifted "domes", making all the rocks vulnerable to acquire high thermochemical remanences with high Q values.

7.2. Rock magnetic properties

High temperature thermomagnetic analysis of the Granophyre samples shows the presence of single domain magnetite, but yielded no indications of mineralogical changes during the experiment. Some of the magnetically more heterogeneous PT breccias also showed reversible behavior and Hopkinson peaks during high temperature thermomagnetic analysis indicating single domain magnetite, but some of them showed irreversible behavior, hints of maghemitization, and the inversion of maghemite. Perhaps they contained clasts of ABC in them, as (and as mentioned above) ABC samples also show hints of maghemitization. More probably though and because we chose unweathered PT breccia samples with no visible clasts, these larger (titano)magnetite (maghemite) grains could be an original feature of the PT breccias. Scanning electron microscope studies revealed the presence of such larger altered (titano)magnetite grains in both PT breccia and Granophyre samples. Hysteresis properties showed that the PT breccia and Granophyre samples fall into the PSD region, which indicates that the remanence carriers may be a mixture of grain sizes between SD and MD.

High temperature thermomagnetic analyses show that many ABC samples have reversible heating and cooling curves, which indicates that no mineralogical changes take place during treatment. Such samples show mainly Curie points in the range from 540 to 580 °C. This suggests (titano)magnetite of variable Ti-content, and this could be verified by scanning electron microscope analysis. The result is consistent with earlier reports (Carporzen et al., 2006; Hart et al., 1995, 2000). However, some of the ABC samples that were studied by us show a strong irreversibility of heating and cooling curves, indicating that mineralogical changes (oxidation) take place during the experiment. Many samples show hints of maghemitization during the experiment, followed by inversion of maghemite. According to their hysteresis properties, the ABC rocks from the Vredefort Dome fall mainly into the PSD region, which may indicate that the magnetite as the remanence carrier involves a mixture of grain sizes between SD and MD. The Lowrie–Fuller test (Lowrie and Fuller, 1971) verified the presence of multi-domain and bimodal magnetic carriers in these samples. ABC samples from the Johannesburg Dome fall either into the SD or into the MD regions.

Carporzen et al. (2006), Cloete et al. (1999) and Hart et al. (2000) also carried out rock magnetic studies. Their results showed that there are two types of magnetite in the ABC samples. They identified the original pre-impact magnetite (stoichiometric) to be larger, close to multi-domain size, and the impact-related magnetite to be a smaller (non-stoichiometric), single-domain size. Based on thermomagnetic analysis of these different magnetite types, Carporzen et al. (2006) proposed temperatures in the core of the Vredefort structure never rose above 550–600 °C for more than 3 min or above 500 °C for 1 h. They obtained two Verwey transitions for Vredefort basement rocks. The first one was at 102 K, which they related to impact-generated non-stoichiometric magnetite, and the second one at 124 K, which they related to pre-impact stoichiometric magnetite. PT breccias showed single Verwey transitions between 94 and 110 K in their study. They noted that when heating separate magnetite fractions above 700 °C, the Verwey transition temperature diminished in large grains and rose or completely disappeared in small grains.

In our low-temperature thermomagnetic analyses we noted that heating some ABC whole-rock samples up to 700 °C does not lead to the disappearance of the higher-temperature transition; instead the lower Verwey transition disappeared. Nor did the heating shift the higher Verwey transition to lower temperatures (Fig. 6). We observed that the value of susceptibility during the second low-temperature treatment is somewhat increased compared to the first low-temperature treatment. It increases relatively more in the samples showing stronger maghemitization during the high-temperature (above room temperature) part of the experiment. Interestingly, we noted two Verwey transitions in one PT breccia sample (Fig. 5b), where we expected to see only one lower temperature Verwey transition. We chose a homogeneous, fresh-looking part of the PT breccia sample for the measurements, and we believe that this observed phenomenon is an original feature of PT breccia and that there may be two types (stoichiometric and non-stoichiometric) of magnetite grains formed, which could both be related to the impact event. The distinction of pre- and syn-impact magnetite grains is not so simple. Moreover, scanning electron microscopy revealed the presence of larger (~50 μm), altered (titano)magnetite grains in both Granophyre and PT breccia samples. This larger magnetite would be more vulnerable to alteration than smaller magnetite. If this is true, then the idea that one may determine impact-related temperatures in the structure based on the heating of magnetite (Carporzen et al., 2006) is not straightforward.

In addition, the fact that some ABC samples with normal NRM intensity values show an impact-related magnetization component carried by magnetite supports the idea that the impact structure experienced at least temperatures needed to reset the magnetization direction of magnetite (~580 °C). Moreover it is clear from the impact model of Ivanov (2005) that the center of the core (minimum radius 15 km) experienced temperatures in excess of 700 °C, and that would have survived for tens of thousands to hundreds of thousands of years after the impact in such originally deeply buried rocks. In addition, metamorphic evidence for post-shock temperatures in excess of 500 °C have been established across the core of the Vredefort Dome (Gibson et al., 1997; Gibson, 2002; Gibson and Reimold, 2005, 2008; see also Reimold et al., 2008).

7.3. Magnetization of impactite samples

A stable hard-coercivity and high-temperature, characteristic remanence component ($D = 18.3°$, $I = 54.8°$, $\alpha_{95} = 8.1°$) of Vredefort impactites was separated from 16 samples (from 56 specimens). As the 95%-error circle of this ChRM component encloses the directions or overlaps with the 95%-error circles obtained in the earlier studies by Carporzen et al. (2006), Hargraves (1970), and Hart et al. (1995), our result is consistent with these studies (Fig. 13 and Table 3).

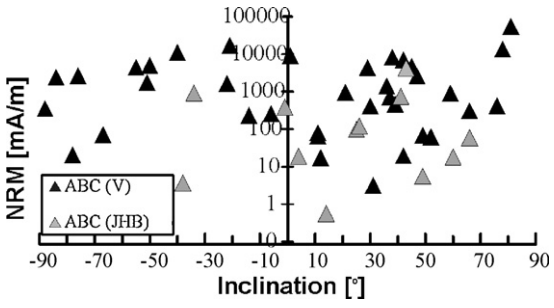


Fig. 13. Mean direction of magnetization of impactite samples from Vredefort Dome with the circle of 95% confidence of directions. Legend as listed in Tables 2 and 3.

Table 3
Palaeomagnetic results for the Vredefort impactites, secondary impact related component of magnetization in Archean basement samples and secondary ca. 1.1 Ga component of magnetization.

(B/)N/n	D (°)	I (°)	k	α_{95} (°)	Plat (°N)	Plon (°E)	A_{95} (°)	S (°)	Comment
Impactites									
12*/16/56	18.3	54.8	30.0	8.1	25.1	43.5	10.6	19.2	This work
6*/47	25.3	55.9	16.8	16.8	22.2	49.2	20.4		Hart et al. (1995)
4*/12	0.0	60.6	62.0	11.8	21.4	27.1	15.7		Hargraves (1970)
2*/31/49	22.9	57.3			21.6	46.6			Carporzen et al. (2005)
Mean pole									
4*/166	18.3	57.5	189.4	5.6	22.8	41.6	10.6		This work
ABC-secondary									
2*/4/14	2.1	62.5	163.5	7.2	19.2	28.9	8.8		Impactite-type component in ABC, this work
ca. 1.1 Ga overprint									
12*/30	159.6	−5.4	5.2	21.1	−54.4	170.6	15.0	35.5	This work

N/n, number of samples/specimens; B in case of this work denotes sites; in case of Carporzen et al. (2005) as number of rock types; *, the statistical level used for mean calculations; D and I, the declination and inclination of the remanent magnetization; k, Fisher's (1953) precision parameter; α_{95}, radius of circle of 95% confidence of direction; Plat and Plong, the latitude and longitude for the paleomagnetic pole; A_{95}, radius of the circle of 95% confidence of the pole; S, the calculated angular standard deviation between poles (Merrill and McElhinny, 1983). ABC Archean Basement Complex.

A similar component with the direction of $D = 2.1°$, $I = 62.5°$ ($\alpha_{95} = 7.2°$; 4 samples and 14 specimens, Table 2) is also seen for some ABC clasts selectively sampled from PT breccia occurrences. These samples show normal NRM intensity values. Most probably this overprint is partly responsible of shifting inclinations of NRM of ABC samples towards shallower values (Fig. 14). A rock magnetic study shows that magnetite is the carrier of the observed magnetization. Hart et al. (1995) also separated a similar impact-related component ($D = 25°$, $I = 56°$, $\alpha_{95} = 16°$) from ABC samples. The fact that an impactite-type direction is separated also from the target rocks indicates that the magnetization carried by impactites is of primary origin and that the heat in the structure at and after impact rose above the lock-in temperature of magnetization of magnetite. A similar component has been seen in various lithologies from the Kaapvaal Craton (e.g., Hart et al., 1995; Jackson, 1982; Maré et al., 2006; Morgan and Briden, 1981; Layer, 1986) and it is interpreted to be related to known ca. 2.0 Ga events (intrusion of Bushveld Complex and/or the Vredefort impact event).

7.3.1. Secular variation

In order to study the continental drift of the Kaapvaal Craton during the Mesoproterozoic the paleomagnetic pole (25.1°N, 43.5°E, $A_{95} = 10.6°$) was calculated for the characteristic remanence component of Vredefort impactites (Tables 2 and 3 and Fig. 15). For this study several Granophyre and PT breccia occurrences were sampled, but the number of the eventually accepted samples is rather small, which may result in the fact that secular variation of the geomagnetic field has not been averaged out (Buchan and Halls, 1990).

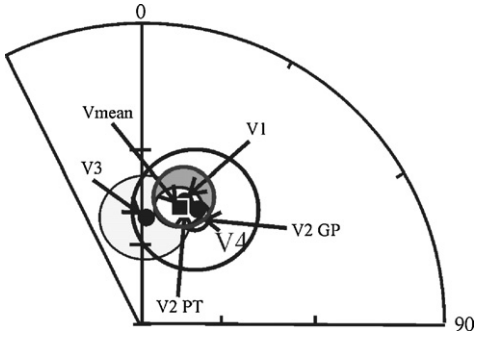


Fig. 14. Natural remanent magnetization vs. inclination of the Archean Basement Core samples from the Vredefort and Johannesburg domes.

If the secular variation is not averaged out, it could cause additional error in the estimation of pole position (Buchan and Halls, 1990). The difference between virtual geomagnetic poles (VGPs) and the paleomagnetic poles can be as much as 15–20° (e.g., McElhinny and McFadden, 2000).

We calculated the angular dispersion (i.e., between-site dispersion, s) of the paleomagnetic poles (e.g., Merrill and McElhinny, 1983) at the site paleolatitude. The s-value was calculated from K (dispersion of poles), and upper and lower limits of s were calculated using the method by Cox (1969). Impactites show a s value of $19.2° + 5.8°/-3.6°$ (Fig. 16) for the dispersion. This was compared with theoretical dispersion, for example with the model G by Merrill and McElhinny (1983) on the paleolatitude of the sampling site (35°). Even though the lower limit falls onto the model G curve, the mean falls well above the SV-model curve of Merrill and McElhinny (1983). This suggests that sampling for this study may not adequately average out the secular variation and use of such pole for paleoreconstructions would be only speculative.

7.3.2. Implications for the Kaapvaal Craton's apparent polar wander path

The paleomagnetic pole (25.1°N, 43.5°E, and $A_{95} = 10.6°$) of the ChRM component of the impactites, as determined by this work, was combined with other data from previous work (Table 3) and the mean pole (22.8°N, 41.6°E, and $A_{95} = 10.6°$) was plotted on the Kaapvaal Craton's apparent polar wander path (APWP) (Fig. 15 and Table 3). This mean pole falls onto the Paleoproterozoic part of the Kaapvaal Craton APWP (see Table 4 for poles and references), which is rather poorly, but somewhat better defined than the Archean part of this APWP. Extensive discussion about the quality of the Archean and Paleoproterozoic poles used for the APWP of the Kaapvaal Craton includes the recent work by Gose et al. (2006), Evans et al. (2002), de Kock et al. (2006), and Strik et al. (2007). An error circle of 95% (A_{95}) for the mean pole overlaps with A_{95} circles for the Bushveld (combined main and upper zone 2061 ± 27 Ma, Evans et al., 2002), Lower Swaershoek Formation (2054 ± 4 Ma, de Kock et al., 2006), and Phalaborwa 1 (2060.5 ± 0.5 Ma, Morgan and Briden, 1981; see also Verwoerd and du Toit, 2006) poles. Statistically it is impossible to distinguish between these poles (Fig. 15 and Table 4). However, the Vredefort pole clearly plots towards the younger (southern) part of the Kaapvaal Craton APWP, in comparison with the Lower Swaershoek and Phalaborwa 1 poles. The Bushveld poles occupy a more southerly part of the APWP compared to the Vredefort poles. Hanson et al. (2004a) suggested that parts of the

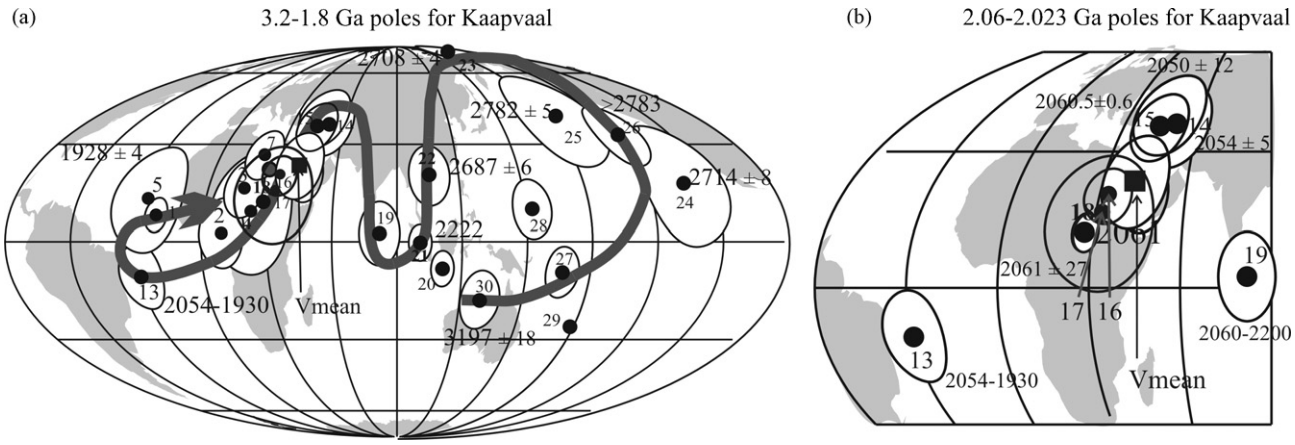


Fig. 15. (a) Mean paleomagnetic pole for Vredefort impactites with its 95% confidence circle (12 in Table 4) plotted among other Archean and Paleoproterozoic paleomagnetic poles for the Kaapvaal Craton (Table 4). (b) Zooming to the 2.06–2.023 Ga poles of Kaapvaal Craton.

Bushveld Complex were remagnetized due to widespread thermal disturbance or fluid migration having occured at the same time as emplacement of the 1.88–1.87 Ga dolerite sills and coeval Soutpansberg mafic rocks (see Maré et al., 2006). On the other hand, the exact timing of magnetic remanence acquisition in the Bushveld Complex is unknown and particularly difficult to constrain, as it is possible that these processes did not occur simultaneously in different parts of this large intrusion (e.g., de Kock et al., 2006), which may result in a mean pole with no well constrained age. Furthermore, Robb et al. (2000) presented evidence for long-lived hydrothermal fluid events (continuing up to 100 Ma after intrusion) that might have affected the Bushveld Complex. de Kock et al. (2006) postulated that the pole of the Lower Swaershoek Formation (36.5°N, 51.3°E, A_{95} = 10.9°) would be a better representation of the Kaapvaal Craton's position during the closing stages of the intrusion of the Bushveld Complex than that provided by the existing paleomagnetic poles from the Bushveld Complex itself. This discussion elucidates, however, that the Kaapvaal Craton could not have experienced significant plate movement between 2061 Ma (Bushveld Complex emplacement) and 2023 Ma (Vredefort impact event) ago. The quality and age of the poles which are younger than the Vredefort poles are questionable, except the pole for Waterberg–Soutpansberg dolerites, which have precise U–Pb ages and the data for which produced a positive baked contact test (Hanson et al., 2004a,b; Gose et al., 2006), but it seems that the APWP of the Kaapvaal Craton is shifted towards the southwest (Fig. 15 and Table 4).

7.4. Magnetization of Archean basement samples

Magnetization of samples from the Archean basement complex of the Vredefort Dome is complex due to high NRM values, as has been discussed above. As the data lack good quality, we are essentially not able to draw any further conclusions on the origin of the observed magnetization. Despite this, we were able to separate an impactite-type component as discussed in Section 7.3, as well as an overprint component, which is interpreted to present younger, ca. 1.1 Ga magnetization, as will be discussed in Section 7.6.

7.5. Alkaline and dioritic intrusions

As an overall result from the analysis of a small number of samples from the studied alkaline and dioritic intrusions (Rietfontein, Lindeques Drift, Roodekraal and Schurwedraai) we may postulate that in situ magnetization directions (Fig. 12a) of these intrusions scatter, and samples show high Q values, which indicates that their magnetization is somewhat distorted. This contradicts the results of Hargraves (1970, 1987) in so far as we are not able to reconstruct the collar overturning, which, however, may be due to the low number of samples analysed so far and/or the high Q values of our samples.

7.6. Younger ca. 1.1 Ga overprint?

Despite the fact that we were not able to see a ca. 1.1 Ga shallow northerly or shallow southerly direction in the data for the Anna's Rust Sheet gabbro samples, we observed such a direction of magnetization (D = 160°, I = −5°, α_{95} = 21°) in some of the impactite and ABC samples (Fig. 11). This magnetic component was seen in samples from both the Vredefort and Johannesburg domes, and such a trend was also reported in the unpublished Honours reports by Jackson (1982) and McDonald and Andersen (1973). Jackson (1982) concentrated on Archean basement granitoids, whereas McDonald and Andersen (1973) studied several gabbro, wehrlite, pyroxenite and granulite sites.

This direction is similar to the known ca. 1.1 Ga direction observed elsewhere in the Kaapvaal Craton (e.g., Umkondo province; Hanson et al., 2004b). We interpret this direction as an overprint caused by ca. 1.1 Ga geological events, known to be widespread on the Kaapvaal Craton, e.g., the emplacement of the Umkondo large igneous province (Hanson et al., 2004b). Moreover, there is local evidence of ca. 1.1 Ga geological activity (intrusion of the Anna's Rust Sheet involving a series of intrusives occurring throughout the extent of the Vredefort Dome; Reimold et al., 2000) on the Vredefort Dome, which could have

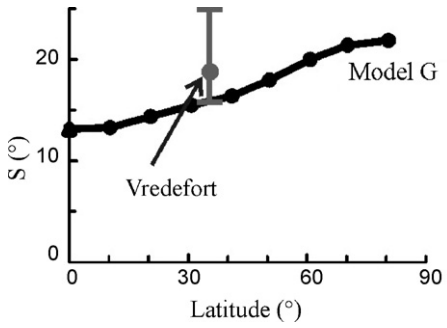


Fig. 16. Calculated between site dispersion of paleomagnetic pole for Vredefort impactites is plotted on the mean curve (model G) for secular variation after Merrill and McElhinny (1983).

Table 4
Paleomagnetic poles used for Kaapvaal Craton's apparent polar wander path.

Formation (no. refers pole in Fig. 15)	Age (Ma)	Plat (°N)	Plon (°E)	dp (°)	dm (°)	References
Mashonaland Sills (1)	1830 ± 230	7.6	338.2	5.1	5.1	McElhinny and Opdyke (1964) and Bates and Jones (1996); see Evans et al. (2002)
Sand River Dykes (2)	1876 ± 68?	2.3	9.1	10.3	10.3	Morgan (1985) and Evans et al. (2002) (recalculation)
Post Waterberg Dolerite (3)	1879–1872	17.4	17.2	8.2	8.2	Gose et al. (2006)
Post Bushveld Dolerites (4)	~1900 (or 1649 ± 10)	8.7	22	18.6	20.6	Letts et al. (2005)
Hartley lava (5)	1928 ± 4	12.5	332.8	16	16	Evans et al. (2002)
Witwatersrand overprint (6)	1945 ± 40?	19.1	45.6	7.8	7.8	Layer et al. (1988)
Limpopo metamorphism "A" (7)	1950–1980?	26.1	22.3	7.9	10.3	Morgan (1985) and Buick et al. (2006)
Vredefort (8, V1)	2023 ± 4	25.1	43.5	8.1	11.4	This work
Vredefort (9, V2)	2023 ± 4	21.6	46.6	6.2	6.2	Carporzen et al. (2005); this work
Vredefort (10, V3)	2023 ± 4	21.4	27.1	13.7	17.9	Hargraves (1970)
Vredefort (11, V4)	2023 ± 4	22.2	49.1	17.3	24.1	Hart et al. (1995)
Vredefort mean (12, V_{mean})	2023 ± 4	22.8	41.6	10.6	10.6	This work
Upper Swaershoek/Alma Formations (13)	2054–1930	−10.5	330.4	9.8	9.8	de Kock et al. (2006)
Lower Swaershoek Formation (14)	2054 ± 4	36.5	51.3	10.9	10.9	de Kock et al. (2006)
Phalaborwa 1 (15)	2060.5 ± 0.6	35.9	44.8	6.9	10.5	Morgan and Briden (1981)
Bushveld middle complex (16)	2060 ± 2	20.0	33.0	5.0	5.0	Hattingh (1989)
Bushveld upper complex (17)	2061 ± 27	16.0	32.0	11.0	11.0	Hattingh (1989, 1999)
Bushveld main and upper zones (18)	2061 ± 27	11.5	27.2	4	4	Evans et al. (2002) (recalculation)
Basal Gamagava/Mapedi (19)	2060–2200	2.2	81.9	7.2	11.5	Evans et al. (2002)
Mamatwan-type Ore Complex 1 (20)	~2200?	−8.1	111.1	5.6	11.1	Evans et al. (2001)
Ongeluk Lavas (21)	2222 ± 13	−0.5	100.7	5.3	5.3	Evans et al. (1997)
Mbabane Pluton (22)	2687 ± 6	−19.7	285.7	9.4	9.4	Layer et al. (1989a,b)
Allanridge Basalt (23)	2708 ± 4	75	180.1	8.8	8.8	Strik et al. (2007)
Westonara Basalt (24)	2714 ± 8	17.1	227.9	18.8	18.8	Strik et al. (2007)
Derdepoort Basalt (25)	2782 ± 5	39.6	184.7	17.5	17.5	Wingate (1998)
Modipe Gabbro (26)	>2783	32.8	210.8	10.4	10.4	Evans and McElhinny (1966)
Usushwana Complex (27)	2875 ± 40	−9.2	167	7.6	7.6	Layer et al. (1988)
Agatha Basalt (28)	2940 ± 22?	9.4	153	8.9	8.9	Strik et al. (2007)
Nhlebela Basalt (29)	2984 ± 2.6?	−25.9	178.4			Strik et al. (2007)
Nelshooste Pluton (30)	3197 ± 18	−17.6	129.8	9	9	Layer et al. (1988)

Plat and Plon, latitude and longitude of paleomagentic pole; dm and dp, the confidence ovals of the pole; Age, the best age for the pole.

been the reason for the observed magnetic overprint on the studied samples.

8. Conclusions

The main result from this study is the stable hard coercivity and the high temperature ChRM component ($D = 18.3°$, $I = 54.8°$, $\alpha_{95} = 8.1°$, 16 samples, 56 specimens) of Vredefort impactites, which is carried by magnetite. Such a direction indicates a pole position of 25.1°N, 43.5°E ($A_{95} = 10.6°$), which takes the Kaapvaal Craton on the paleolatitude of 35°. Owing to the small number of samples that yielded sufficiently meaningful results, it is possible that the secular variation has not been sufficiently averaged out and the pole represents only a virtual geomagnetic pole, indicating that its use for paleoreconstructions may be speculative. However, the observed direction of magnetization and pole is in good agreement with previous results by Carporzen et al. (2005), Hargraves (1970) and Hart et al. (1995, 2000). Four Archean basement samples with normal intensity values of natural remanent magnetization show a similar impactite-type component with the mean direction of $D = 2.1°$, $I = 62.5°$ ($\alpha_{95} = 7.2°$; 4 samples and 14 specimens). The mean pole for Vredefort impactites falls on the Paleoproterozoic part of the Kaapvaal Craton's APWP.

An important result from petrophysical and rock magnetic studies is that high Q values, also reported by previous studies (Hart et al., 1995, 2000; Carporzen et al., 2005, 2006), are now also seen in the Johannesburg Dome, which rules out the direct Vredefort impact connection as an explanation. This is also supported by the fact that there is no correlation between hysteresis data and high Q values of the basement rocks, as would be expected if the ultrafine particles in the planar deformation features were the cause of the high Q values. We excluded lightning as a cause of all the observed high Q values. The explanation for the high Q values seems to be related to the elevated temperatures and likely fluid circulation within the two domes that had the potential to make the rocks vulnerable to acquire high thermochemical remanences.

Low temperature vs. susceptibility measurements (Verwey transitions) indicate that the distinction of pre- and syn-impact magnetite grains is not so straightforward as postulated by Carporzen et al. (2006). In addition, the fact that some ABC samples with normal NRM intensity values show an impact-related magnetization component carried by magnetite supports the idea that the impact structure experienced at least temperatures needed to reset the magnetization direction of magnetite (~580 °C).

Detailed analysis of all rocks, including the Vredefort impactites, yielded occasionally distinct great circle paths either towards a shallow north or a shallow south direction. As this direction is also isolated from the baked zone of the ARS-gabbro intrusion in the northern part of the collar of the Vredefort structure, we interpret it as caused by the emplacement of the Umkondo large igneous province.

As the data for samples from the Rietfontein, Schurwedraai, Lideques Drift, and Roodekraal intrusions are quite sparse, low quality and scattered, we are not able to draw any conclusion about their meaning, at this time.

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